

Basics of Genetics Educational Workshop

Educational Workshop - 10th July 2026



INTRODUCTION TO THE COMMAND LINE

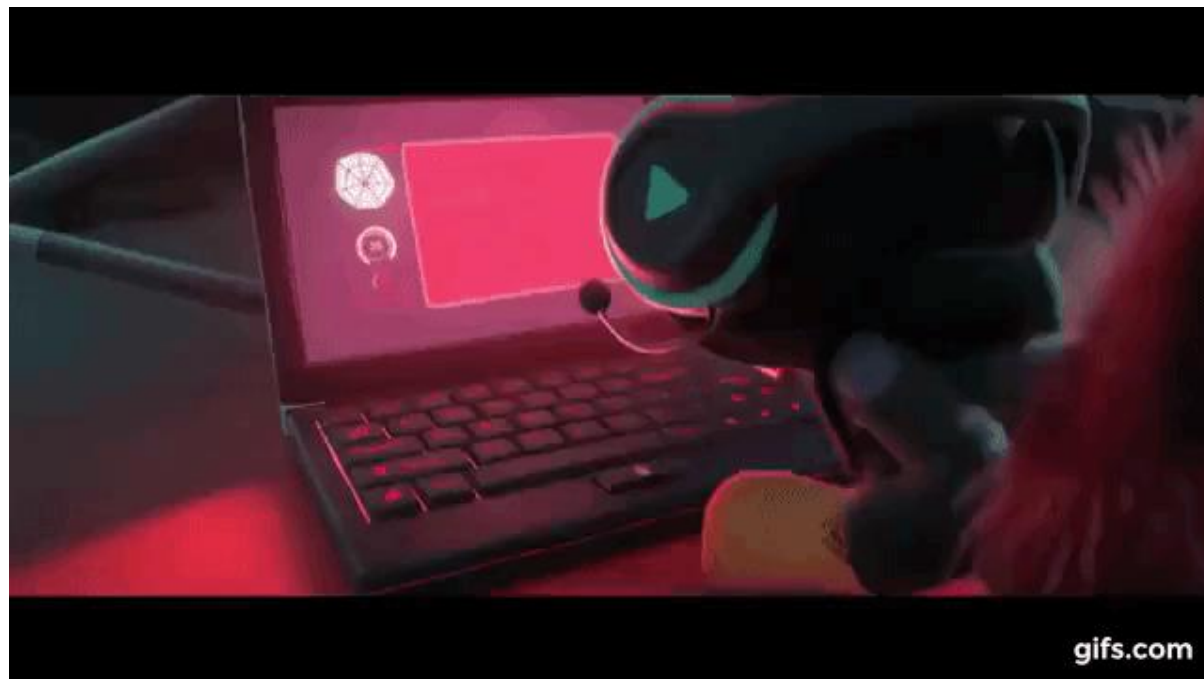
CORTESY: DR DAVID CASH

- Why use the command line?
- Command grammar
- Common commands
- Files and Directories
- Big data

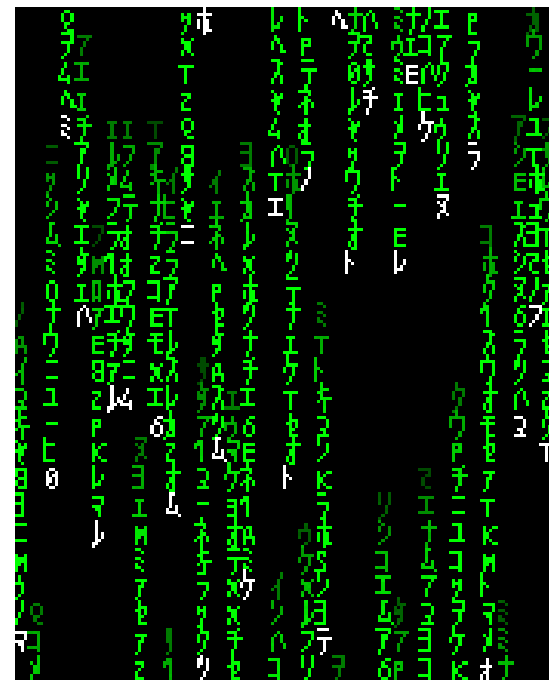
By the end of this section, you should be able to:

- Understand the benefits of the command line and the basic grammar of a shell command
- Run some basic commands
- Navigate through file and directory structures

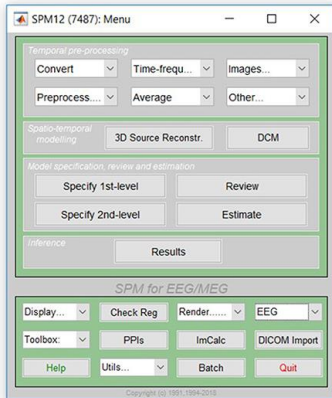
WHAT THE COMMAND LINE IS NOT!



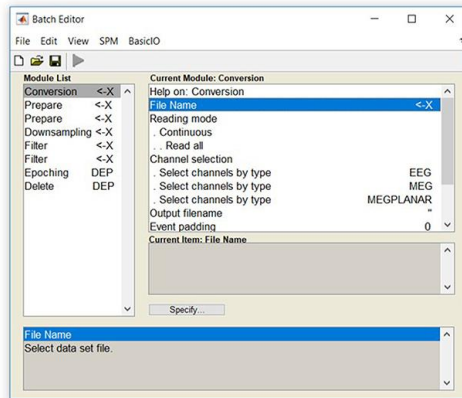
gifs.com



COMMAND LINE INTERFACE (CLI) VS GRAPHICAL USER INTERFACE (GUI)



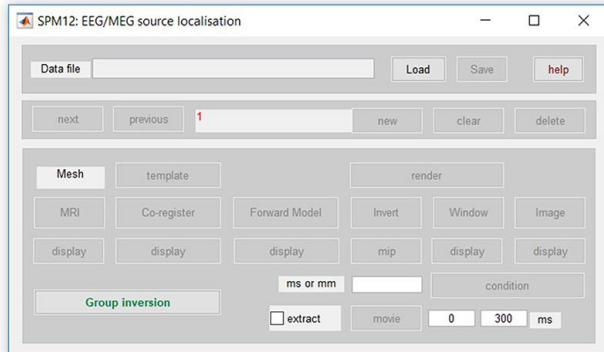
A



B

```
davecash — -zsh — 80x24

Last login: Wed May 18 22:30:12 on ttys002
(base) davecash@Davids-MacBook-Pro ~ % pwd
/Users/davecash
(base) davecash@Davids-MacBook-Pro ~ % ls
15-016-12                               Public
Applications                             PycharmProjects
Data                                     University College London
Desktop                                 VirtualBox VMs
Documents                             Zotero
Downloads                             anaconda3
Library                               install
Movies                                matlab
Music                                 src
OneDrive - University College London   test_out2.png
Pictures
(base) davecash@Davids-MacBook-Pro ~ %
```



C

Graphical User Interface

- ✓ Simple
- ✓ Intuitive
- ✗ Can restrict functionality
- ✗ Repetitive work = more mouse clicks, more errors
- ✗ More difficult to automate

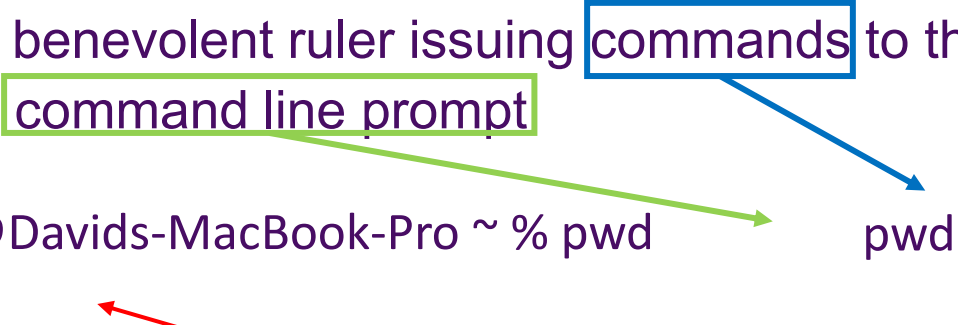
Command Line

- ✓ Easier to automate many tasks
- ✓ More flexible
- ✓ Can piece multiple programs together
- ✓ Efficient when communicating remotely
- ✗ Learning curve
- ✗ No common options provided
- ✗ Feedback is not always clear or helpful

THE COMMAND LINE IS A SIMPLE LANGUAGE

- The most common command line interface we use is called the shell, which is available in some form on most computers
- Consider yourself the benevolent ruler issuing **commands** to the computer through the **command line prompt**

(base) davecash@Davids-MacBook-Pro ~ % pwd
/Users/davecash



- Your humble servant, the computer, will try its best to fulfill your command and, in most cases, give a **response** including if the command was not successful

“Copy the file called myfile into a new file called yourfile and give me a detailed (verbose) response”

`cp -v myfile yourfile`

Verb: the name of the command itself – this is mandatory

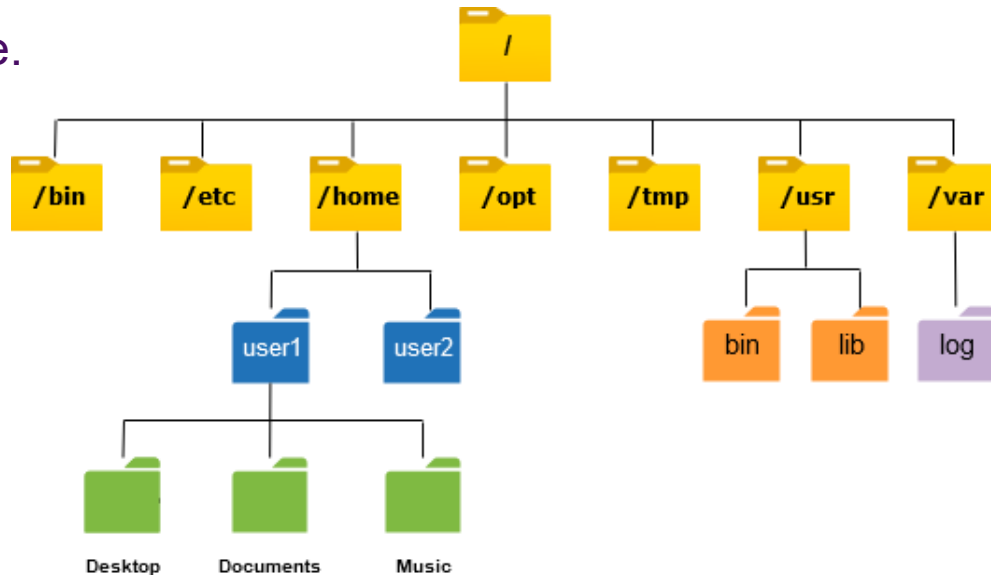
Object(s): what the command is acting upon (files, text, numbers, etc)

Adjective(s)/Adverb(s): How to perform the command. These are often called command line options. They are often prefaced by a dash

Slang: Shortcuts to save keystrokes

Command	Name	Function	Usage
man	Manual	HELP!	man cd
pwd	Print working directory	Where am I?	pwd
mkdir	Make directory	Create a new directory	mkdir dir1
cd	Change directory	Go to the following location	cd dir1 cd /home/dir1
ls	List	Shows what is inside a directory	ls dir1
cp	Copy	Copies a file to a new location	cp source destination
mv	Move	Moves a file to a new location	mv source destination
rm	Remove	Deletes a file or a directory	rm dir1/bad_file
cat	Concatenate	Prints out the contents of a file	cat results.txt
emacs	Text editor	Edit plain text files	emacs dir1/inputs.txt

- The file system is an upside-down tree. At the top is the **root directory**
- Your starting location is your **home directory**, which is your own private space.
- Every file and directory has an **owner**, who controls the **permissions** of who can access it
- Files often have an extension at the end (.sh, .txt, .doc, .nii) to indicate the file type



- A path is the term typically used to describe the unique location of a directory or file.
- Paths can either be **absolute** or **relative**

Absolute from anywhere

Windows: C:\home\cash\work

Linux: /home/cash/work



Relative from home subdirectory “play”

Windows: ..\work

Linux: ../work

← from San Diego International Airport
to Hilton San Diego Bayfront, 1 Park Blv...

10 min (4.3 miles)

via N Harbor Dr

Fastest route now due to traffic conditions

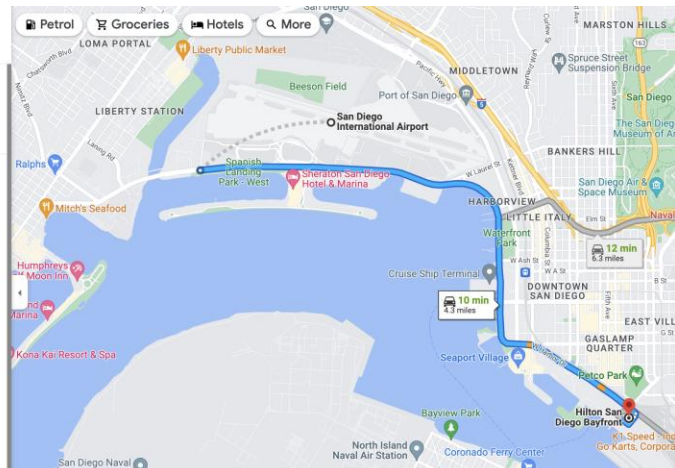
San Diego International Airport
3225 N Harbor Dr, San Diego, CA 92101,
United States

↑ Head east on N Harbor Dr towards
McCain Rd
4.2 mi

→ Turn right onto Switzer St
128 ft

↑ Continue straight onto Gull Street
0.1 mi

Hilton San Diego Bayfront
1 Park Blvd, San Diego, CA 92101, United
States



- Windows uses \ while Linux uses /
- The root of the tree in Windows is a letter (C:, D:), there is only one root in Linux which is /
- Shortcuts for special directories: . (Current Directory), .. (Up one level/parent directory) and ~ (home directory)

- Outside of the standard commands that are built into the Shell, researchers and software developers create new programs that can be called on the command line
- These commands follow the same grammatical structure as the standard commands but will perform various image analysis tasks
- Most come with a --help option (or similar) that will print out an explanation of how to use the command
- Now let's use these commands in the following lectures to process and visualize different types of brain images